

Isaac Nathan da Silva Barbosa

Full-Stack AI Engineer · Computer Engineering · AI · Cloud · IoT

PWD — ASD (ICD-11: 6A02.2) + ADHD (ICD-11: 6A05.2)

isaacnathandasilva@gmail.com · +55 (31) 99441-7786 · Mariana, MG — Brazil · linkedin.com/in/isaac-nathan · github.com/xAngryBadger · xangrybadger.github.io/isaac-vitae

PROFILE

Full-Stack AI Engineer · Python · FastAPI · React 19 · Azure Cosmos DB · GPT-4.1 · Flutter · PyTorch · 10 months of professional experience. First Python contact in 2022 at UFOP (Industrial Chemistry) — Thonny IDE, no GPT, extra afternoon classes just to keep learning. Built ForestAI from scratch using Stack Overflow and Thonny, manually annotating hundreds of drone images from Fundação Renova. At Paware, migrated databases to Azure Cosmos DB (Meritage Homes, USA) and architected AI pipelines for HelloSocial (Flux, DALL-E 3, Canva Connect). I learn by solving real problems — from cross-platform MIME types to ReAct agents with schema enforcement.

CORE STACK

Frontend: React 19 · TypeScript · Flutter · Tailwind CSS v4 · Vite · React Native

→ HarpIA frontend, Flora Sensus admin panel, Inovesa institutional site, this portfolio.

Backend: Python · FastAPI · Node.js · Express · Azure Cosmos DB · SQL

→ HarpIA pipeline, AguaQuality IoT backend, HelloSocial (Paware). Python async as production language.

Cloud & Infra: Azure · Docker · Git/GitHub · PocketBase · Linux (CachyOS/Hyprrland)

→ Database migration to Azure Cosmos DB (Meritage Homes, USA). Docker for AI agents. PocketBase as Flora Sensus backend.

AI / ML: GPT-4.1 · PyTorch · DeepForest · OpenCV · Ollama · DALL-E 3 · Flux · scikit-learn

→ 7+ AI models integrated in HarpIA. ReAct agent with Ollama in Fennec. Species detection with PyTorch + DeepForest in ForestAI.

Integrations: Mercado Pago PIX · Canva Connect · Placid API · Pexels API · NVIDIA NIM

→ PIX via Mercado Pago (AguaQuality). Canva + Placid (HelloSocial/Paware). Pexels (HarpIA).

PROFESSIONAL EXPERIENCE

Full-Stack Developer with AI focus

Paware Softwares · Oct 2025 — May 2026

- Migrated legacy datasets from Google Drive to Azure Cosmos DB for Meritage Homes (USA) — pipeline with automated cookie-based extraction, compression, renaming, and injection into panels for WhatsApp agent embedding with strict file size limits. Containerized with Docker for reproducible environments and wrote schema validation layer with automated rollback.
- Resolved cross-platform MIME type issue (Android native vs iPhone requiring octet-stream — tested on my parents' iPhone, no Macbook). This validation layer became the backbone of the final Azure Cosmos DB migration.
- Architected agentic image-generation pipeline for HelloSocial — GPT-4.1 agent with tool calling, up to 10 reasoning iterations, integrating Flux Kontext Pro + DALL-E 3 + Placid/Canva APIs. Fallback to SQLite/local PIL when paid APIs weren't needed.

Robotics Instructor

SuperNerds · Sep 2025 — Oct 2025

- Learned Arduino and LEGO in under 2 weeks and taught robotics to children/teens, connecting theory to real applications.
- Balanced mornings here with evening work at Paware.

ENGINEERING PROJECTS

HarpIA · github.com/xAngryBadger/harpia

Python · GPT-4.1 · DALL-E 3 · Flux 2.0 Pro · Sora · Veo 3.1 · Azure Cosmos DB · SQLite · PIL

- Creative automation engine with 7+ AI models. Autonomous agentic pipeline: GPT-4.1 orchestrates tool calls (image generation, Pexels search, copywriting, design compositing) with schema enforcement and up to 10 reasoning iterations (think → call → observe).
- Lightweight stack by default: SQLite + local PIL renders designs with 8 custom templates, brand colors, and badge overlays; falls back to paid APIs when needed (DALL-E 3, Flux, Sora, Veo 3.1).
- Backend swap: local SQLite for development, Azure Cosmos DB + Blob Storage for production, toggled via environment variable. Cron-ready with file-locking and stuck batch recovery.
- 6,930+ LOC async Python with security tests and zero hardcoded secrets.

Flora Sensus · github.com/xAngryBadger/flora-sensus

Flutter · Dart · Drift/SQLite · React · PocketBase · TypeScript

- Flutter offline-first app for forest inventory with custom sync engine: timestamp-based conflict detection, cascading UUID remapping through FK chain (Propriedade → UT → Parcela → Planta → Foto), and atomic rollback via Drift transactions.
- ~24K LOC — architecture and sync logic built from scratch; code generated with LLM assistance (web) and manually reviewed. 5-tier hierarchy with XLSX/PDF/CSV export.
- React admin panel with auth, photos, reports, and export. Exponential backoff with jitter, auth retry wrapper, ngrok bypass for development.

ForestAI · github.com/xAngryBadger/forestai

PyTorch · DeepForest · OpenCV · scikit-learn · TensorBoard

- Forest species detection and classification with Deep Learning — built from scratch without AI-assisted coding. Stack Overflow + Thonny IDE only.
- Manual annotation of hundreds of drone images from Fundação Renova (bounding boxes), DeepForest/YOLO training on local GPU, stratified splits. TensorBoard curve interpretation — detecting memorization vs generalization. The hard way built the intuition that made every subsequent framework click faster.

Fennec Excel · github.com/xAngryBadger/Sahara-Fenneck

Python · Ollama/qwen2.5 · CustomTkinter · xlwings/COM · PyInstaller · Inno Setup

- Local AI assistant for Excel via ReAct agent (Ollama/qwen2.5). Natural language commands to filter, sort, rename sheets, and manipulate spreadsheets with auto-checkpoint before every change.
- 6+ OAuth integrations (Gmail, Teams, Calendar, Drive, Outlook, Trello) with user confirmation before modifications.
- Hand-crafted visual design with pastel palette and original mascot (Fennec). Native Windows installer (Inno Setup + PyInstaller). Bilingual PT/EN interface.

SRF System · github.com/xAngryBadger/srf-system

Python · pandas · openpyxl · NiceGUI · Rich CLI · unittest

- Operational planning engine for large-scale forest restoration — automatic generation of executive dossiers with crew allocation, territory mapping, and schedule management.
- Tariff and operational cost management. NiceGUI + Rich CLI interface with unit test suite.

MaineCoon · github.com/xAngryBadger/mineral

Node.js · mineflayer · NVIDIA NIM API · Reinforcement Learning · pathfinder

- Minecraft bot with natural language commands via LLM (NVIDIA NIM API) — mine, craft, follow, navigate, and interact through chat. Reinforcement learning module for autonomous behavior.

HelloSocial

Python · FastAPI · Azure OpenAI · Flux · Canva Connect · React · TypeScript

- AI-powered social media post creation and scheduling platform — project at Paware that inspired HarpIA. Image generation pipeline with Flux Kontext Pro and DALL-E 3, copy and template agents.

EDUCATION

Computer Engineering

Cruzeiro do Sul · 2024 — 2029

5th semester — in progress

Chemical Engineering

UFSJ — Campus Alto Paraopeba · 2022 — 2024

Pivot period — away from code, watching AI closely

Industrial Chemistry

UFOP · 2022

Where Python began — Thonny IDE, no GPT

CERTIFICATIONS

- **Python Essentials 1** — Cisco Networking Academy — Programming fundamentals with Python — logic, structures, and first scripts.
- **Python Essentials 2** — Cisco Networking Academy — Advanced Python — OOP, libraries, and certification prep.
- **Data Science Essentials with Python** — Cisco Networking Academy — Data analysis with Pandas and Matplotlib — hands-on, project-based learning.
- **Data Analytics Essentials** — Cisco Networking Academy — Essential analytics tools recognized by the industry.
- **Networking Basics** — Cisco Networking Academy (120h) — Completed during Computer Engineering at Cruzeiro do Sul.
- **Intro to Cybersecurity** — Cisco Networking Academy — Foundation in network security and cyber threats.
- **AI for Process Optimization & Decision-Making** — Escola Virtual Gov · Enap · Serpro (71h) — Program by the Gov AI Nucleus (PBIA) — strategic use of AI in public management and information security.
- **IT Security** — Bradesco Foundation — Complement in corporate infrastructure and data protection.
- **Fluent English** — KUMON (3 years) — Enables technical documentation and meetings with international teams.

Languages: Portuguese — Native · English — Fluent (KUMON, 3 years)